

Python is that language for a growing number of traders and financial institutions.

In this chapter, we're taking our first steps into Python programming. Think of it as learning the alphabet and basic vocabulary of a new language—one that will eventually allow you to communicate sophisticated trading ideas to your computer. We'll explore the fundamental building blocks: variables that store information, data types that categorize it, operators that manipulate it, and input/output operations that let you communicate with your program.

By the end of this chapter, you'll write simple Python code that could form the foundation of a trading system. Let's dive in.

Variables: Your Digital Storage Containers

What Are Variables?

In the financial world, information is everything. Stock prices, trading volumes, economic indicators—these pieces of data drive trading decisions. In Python, we store and manipulate this information using variables.

Think of variables as labeled containers that hold information. Just as a trader might have different folders for different stocks in their research, Python variables let you organize information by giving it names.

Creating a variable in Python is remarkably simple:

```
stock_price = 142.58
```

That's it! We've just created a variable named `stock_price` and stored the value 142.58 in it. Now whenever we refer to `stock_price` in our code, Python knows we're talking about 142.58—at least until we decide to change it.